Ceramic materials for SiO2-MoS2 composites

"what are the possible materials (ceramic ones) that can make a three parts composite with SiO2 and MoS2?" *Generated by LeapSpace*

The clearest ceramic third components supported by the retrieved evidence are *TiO_2* and *$H_2Ti_2O_5$* (titanate nanotubes/whiskers); both appear in three-part systems with *SiO_2* and *MoS_2*. The evidence also shows that **SiO_2 itself is often used as the ceramic matrix or shell around MoS_2**, but that is a binary SiO_2–MoS_2 system rather than a distinct third ceramic component. [95] [96] [97] [98]

Ceramic options that fit a SiO_2 + MoS_2 + third-component composite

Ceramic third component	How it appears with SiO_2 and MoS_2	Why it is a plausible choice
TiO_2	Explicitly used in a **$MoS_2/SiO_2/TiO_2$** ternary heterojunction composite for anti-biofouling and anti-corrosion applications. [95]	TiO_2 is a robust ceramic oxide that can pair with SiO_2 and MoS_2 in a ternary structure. [95]
$H_2Ti_2O_5$ titanate	Used in **$H_2Ti_2O_5$@MoS_2@SiO_2** core-shell nanoparticles. [96]	The titanate core provides a dielectric ceramic base, while MoS_2 and SiO_2 form the active shell/interface structure. [96]
SiO_2 as the ceramic shell/matrix	MoS_2–SiO_2 core-shell and matrix/filler systems are reported, including **SiO_2@MoS_2** fillers and MoS_2/silica coatings. [98] [99]	This supports SiO_2 as a structural ceramic partner, though not as the "third" component if you already count SiO_2 and MoS_2. [98] [99]

What this means in practice

- If you want a **true three-part ceramic composite** with **SiO_2 + MoS_2 + one more ceramic**, the strongest directly supported choice from the evidence is **TiO_2**. [95]
- If you want a **core-shell ceramic framework**, **$H_2Ti_2O_5$** is also supported as the ceramic scaffold with **MoS_2 and SiO_2** added around it. [96]
- The evidence does **not** directly establish a broad list of all possible ceramic third phases; it only supports these specific examples. [95] [96]

Bottom line: for a SiO_2/MoS_2-based ternary ceramic composite, the most defensible ceramic third materials from the retrieved evidence are **TiO_2** and **$H_2Ti_2O_5$ (titanate)**, with **SiO_2** commonly serving as the ceramic matrix or shell rather than the extra third phase. [95] [96] [98]

Confidence level: **Medium** — the evidence directly supports a few concrete ternary or core-shell examples, but it does not provide a comprehensive survey of all ceramic third-component options.

References

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